EAST WEST UNIVERSITY

Department of Computer Science and Engineering

Spring 2026

Course Code: CSE405

Course Title: Computer Networks

Section: 03

Mini Project Report

Submitted by:

Name: Samiha Hasan

ID: 2023-2-60-054

Submitted to:

Dr. Anisur Rahman

Associate Professor

Department of Computer Science and Engineering

Submission Date: 01.05.2026

Design and Implementation of Apex University Campus Network

Introduction

This project focuses on designing and implementing a full-fledged enterprise network for Apex University using Cisco Packet Tracer.

The network was designed to simulate a real university infrastructure consisting of multiple campuses connected through routers using serial links. The project includes dynamic routing, centralized IP allocation, domain name resolution, web hosting and wireless connectivity.

The purpose of this project is to understand practical implementation of:

- Enterprise network design
- Subnetting
- Dynamic routing
- DHCP configuration
- DNS and Web server deployment
- Wireless networking

Project Objective

The objective of this project is to design a complex network for Apex University that fulfills all specified requirements.

The network ensures:

- Connectivity between all campuses
- Automatic IP address allocation
- Access to university website using domain name
- Dynamic routing using OSPF
- Wired and wireless host support
- Scalable network architecture

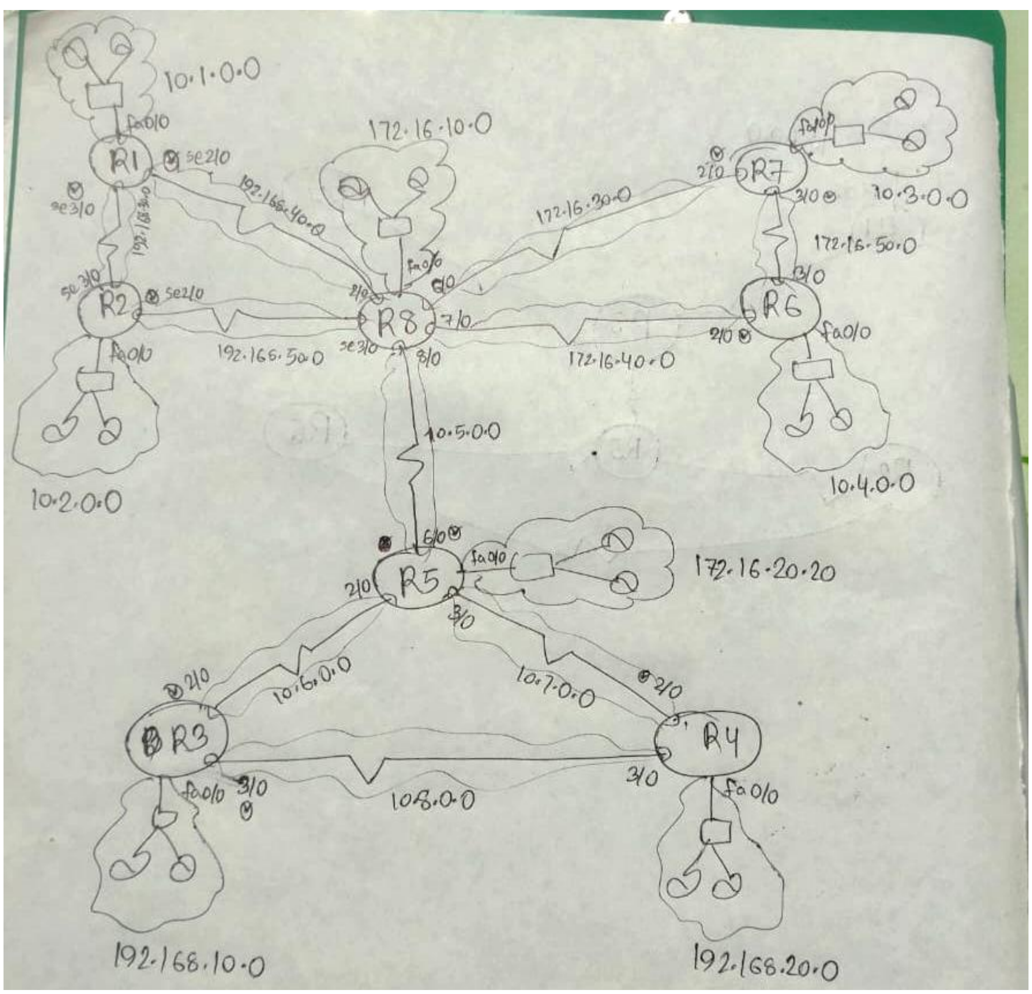
Network Design Overview

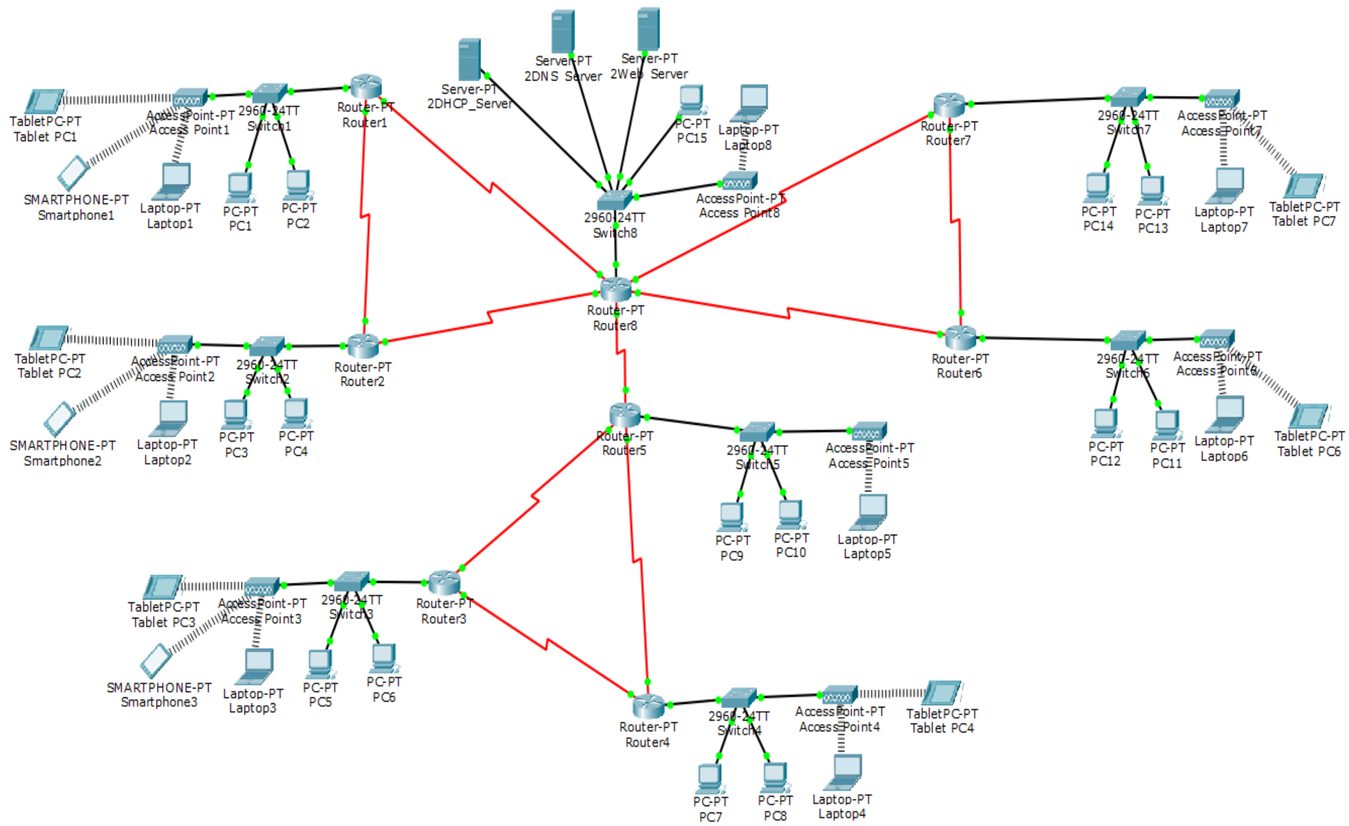
The network consists of:

- 8 Campus Routers
- Multiple Switches
- Wired Hosts
- Wireless Hosts
- Access Points
- DHCP Server
- DNS Server
- Web Server

Each campus represents an individual LAN.

All routers are interconnected using serial connections following the provided topology.





Network Components

Routers

Eight routers were used to connect the eight campuses.

Functions:

- Inter-campus communication
- Packet forwarding
- Dynamic route exchange using OSPF

Switches

Switches were used to connect end devices within each campus.

Functions:

- Local traffic forwarding
- Device connectivity

IP Addressing Scheme

The network uses three IP classes.

Campus	Router LAN IP	Network	IP Class
Campus1	10.1.0.254	10.1.0.0	Class A
Campus2	10.2.0.254	10.2.0.0	Class A
Campus3	192.168.10.254	192.168.10.0	Class C
Campus4	192.168.20.254	192.168.20.0	Class C
Campus5	172.16.20.254	172.16.20.0	Class B
Campus6	10.4.0.254	10.4.0.0	Class A
Campus7	10.3.0.254	10.3.0.0	Class A
Campus8	172.16.10.254	172.16.10.0	Class B

Routing Configuration

Router 1	Router 2
interface fa0/0 ip address 10.1.0.254 255.255.255.0 interface se2/0 ip address 192.168.40.1 255.255.255.0 clock rate 64000 interface se3/0 ip address 192.168.30.1 255.255.255.0 clock rate 64000	interface fa0/0 ip address 10.2.0.254 255.255.255.0 interface se2/0 ip address 192.168.50.1 255.255.255.0 clock rate 64000 interface se3/0 ip address 192.168.30.2 255.255.255.0
Router 3	**Router 4**
interface fa0/0 ip address 192.168.10.254 255.255.255.0 interface se2/0 ip address 10.6.0.1 255.255.255.0 clock rate 64000 interface se3/0 ip address 10.8.0.1 255.255.255.0 clock rate 64000	interface fa0/0 ip address 192.168.20.254 255.255.255.0 interface se2/0 ip address 10.7.0.1 255.255.255.0 clock rate 64000 interface se3/0 ip address 10.8.0.2 255.255.255.0
Router 5	**Router 6**
interface fa0/0	interface fa0/0

ip address 172.16.20.254 255.255.255.0 interface se2/0 ip address 10.6.0.2 255.255.255.0 interface se3/0 ip address 10.7.0.2 255.255.255.0 interface se6/0 ip address 10.5.0.1 255.255.255.0 clock rate 64000	ip address 10.4.0.254 255.255.255.0 interface se2/0 ip address 172.16.40.1 255.255.255.0 clock rate 64000 interface se3/0 ip address 172.16.50.2 255.255.255.0
Router 7	**Router 8**
interface fa0/0 ip address 10.3.0.254 255.255.255.0 interface se2/0 ip address 172.16.30.1 255.255.255.0 clock rate 64000 interface se3/0 ip address 172.16.50.1 255.255.255.0 clock rate 64000	interface fa0/0 ip address 172.16.10.254 255.255.255.0 interface se2/0 ip address 192.168.40.2 255.255.255.0 interface se3/0 ip address 192.168.50.2 255.255.255.0 interface se6/0 ip address 172.16.30.2 255.255.255.0 interface se7/0 ip address 172.16.40.2 255.255.255.0 interface se8/0 ip address 10.5.0.2 255.255.255.0

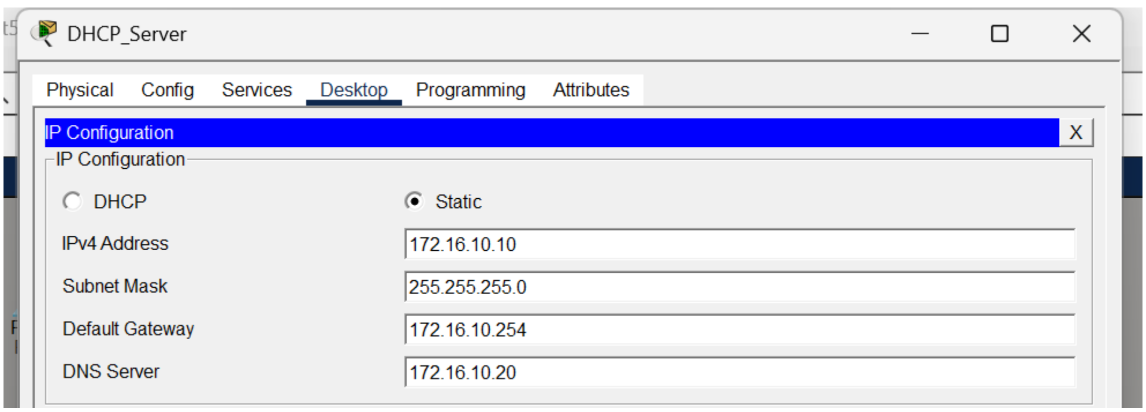
Dynamic routing Configuration

Router 1	**Router 2**
router ospf 1 network 10.1.0.0 0.0.0.255 area 0 network 192.168.40.0 0.0.0.255 area 0 network 192.168.30.0 0.0.0.255 area 0	router ospf 1 network 10.2.0.0 0.0.0.255 area 0 network 192.168.50.0 0.0.0.255 area 0 network 192.168.30.0 0.0.0.255 area 0
Router 3	**Router 4**
router ospf 1 network 192.168.10.0 0.0.0.255 area 0 network 10.6.0.0 0.0.0.255 area 0 network 10.8.0.0 0.0.0.255 area 0	router ospf 1 network 192.168.20.0 0.0.0.255 area 0 network 10.7.0.0 0.0.0.255 area 0 network 10.8.0.0 0.0.0.255 area 0
Router 5	**Router 6**
router ospf 1	router ospf 1

network 172.16.20.0 0.0.0.255 area 0 network 10.6.0.0 0.0.0.255 area 0 network 10.7.0.0 0.0.0.255 area 0 network 10.5.0.0 0.0.0.255 area 0	network 10.4.0.0 0.0.0.255 area 0 network 172.16.40.0 0.0.0.255 area 0 network 172.16.50.0 0.0.0.255 area 0
Router 7	**Router 8**
router ospf 1 network 10.3.0.0 0.0.0.255 area 0 network 172.16.30.0 0.0.0.255 area 0 network 172.16.50.0 0.0.0.255 area 0	router ospf 1 network 172.16.10.0 0.0.0.255 area 0 network 192.168.40.0 0.0.0.255 area 0 network 192.168.50.0 0.0.0.255 area 0 network 172.16.30.0 0.0.0.255 area 0 network 172.16.40.0 0.0.0.255 area 0 network 10.5.0.0 0.0.0.255 area 0

DHCP Server

A centralized DHCP server was configured.



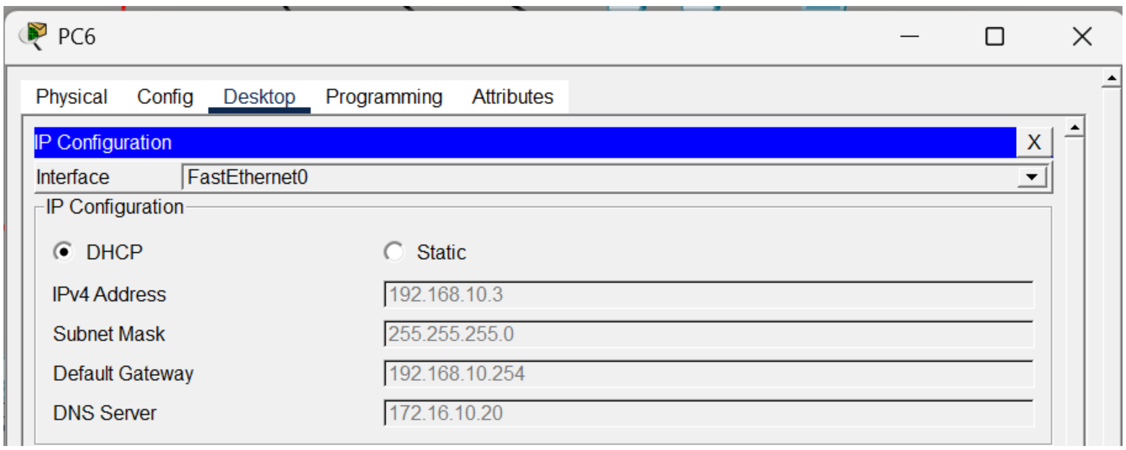
Add | Save | Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
Campus8	172.16.10.254	172.16.10.20	172.16.10.100	255.255.255.0	50	0.0.0.0	0.0.0.0
Campus3	192.168.10.254	172.16.10.20	192.168.10.1	255.255.255.0	50	0.0.0.0	0.0.0.0
Campus7	10.3.0.254	172.16.10.20	10.3.0.1	255.255.255.0	50	0.0.0.0	0.0.0.0
Campus6	10.4.0.254	172.16.10.20	10.4.0.1	255.255.255.0	50	0.0.0.0	0.0.0.0
Campus5	172.16.20.254	172.16.10.20	172.16.20.1	255.255.255.0	50	0.0.0.0	0.0.0.0
Campus4	192.168.20.254	172.16.10.20	192.168.20.1	255.255.255.0	50	0.0.0.0	0.0.0.0
Campus2	10.2.0.254	172.16.10.20	10.2.0.1	255.255.255.0	50	0.0.0.0	0.0.0.0
Campus1	10.1.0.254	172.16.10.20	10.1.0.1	255.255.255.0	50	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	172.16.10.0	255.255.255.0	512	0.0.0.0	0.0.0.0

Functions:

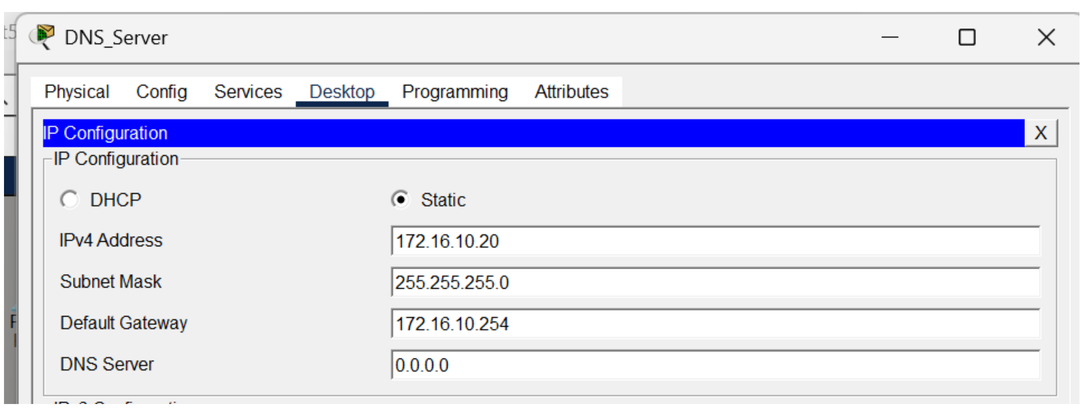
- Automatic IP address allocation
- Subnet mask assignment
- Default gateway assignment
- DNS server assignment

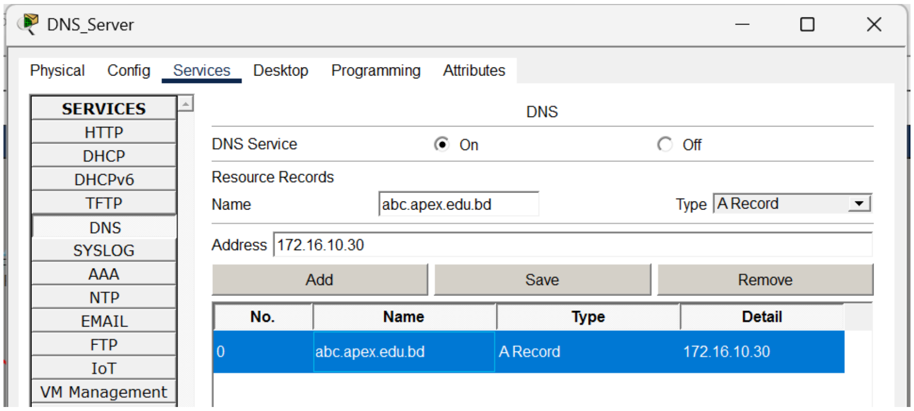
Wired hosts: successfully received IP addresses via DHCP.



DNS Server

This allows users to access the university website using domain name (abc.apex.edu.bd) instead of IP address (172.16.10.30).

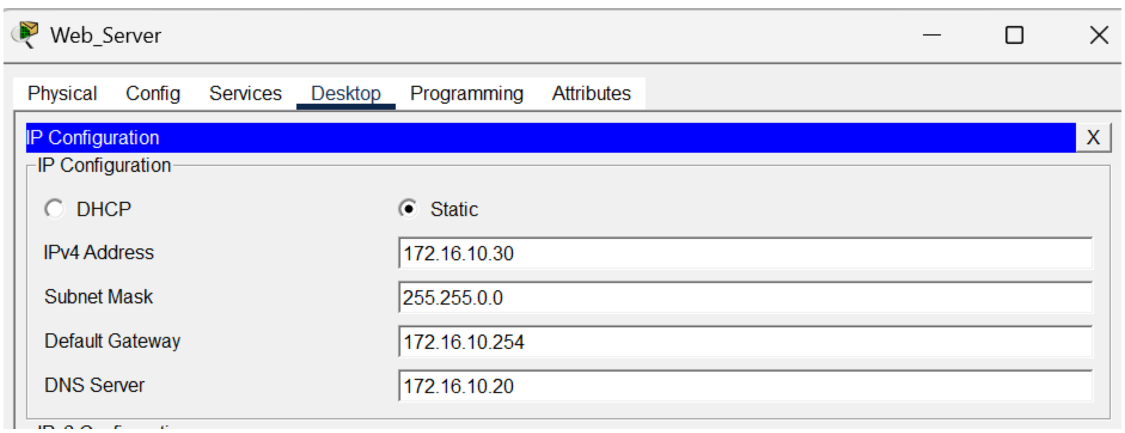


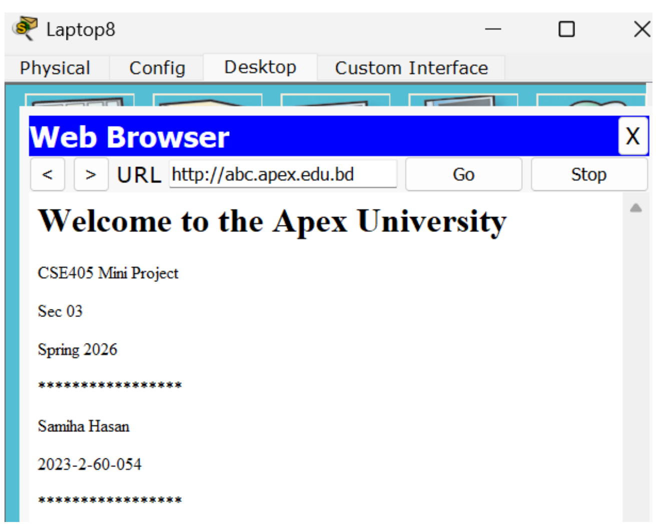


Web Server

HTTP service was enabled to display:

Welcome to the Apex University

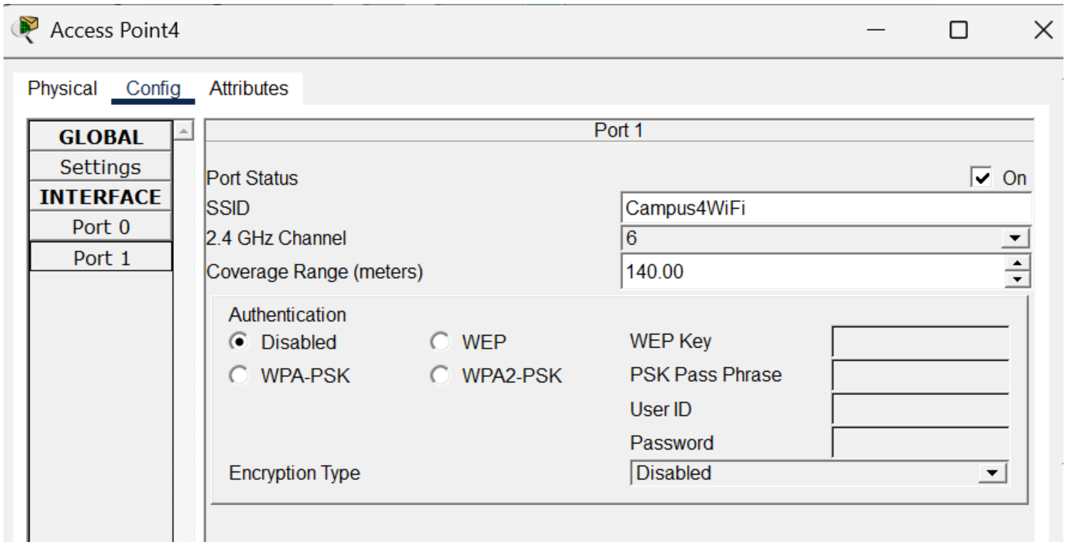




Wireless Configuration

Access Points

Wireless access points were deployed in each campus.



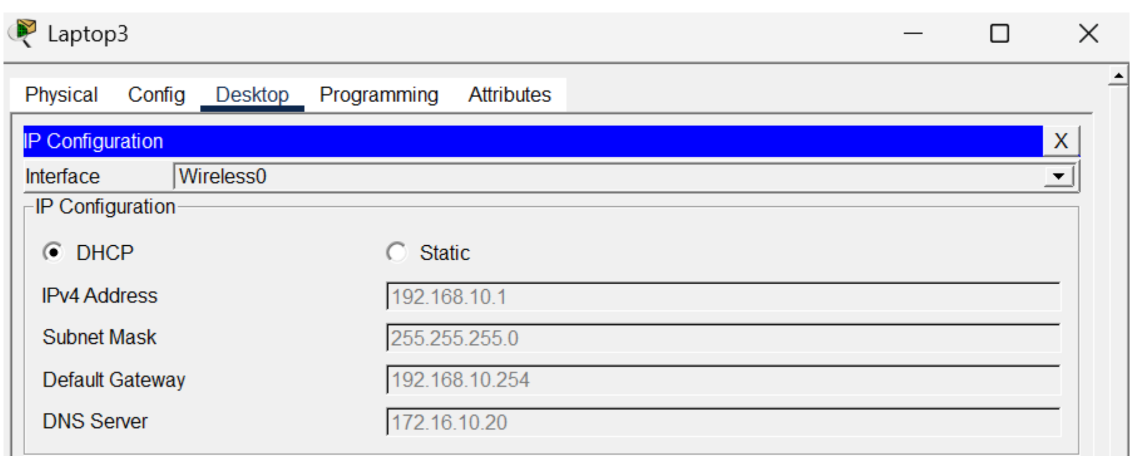
Functions:

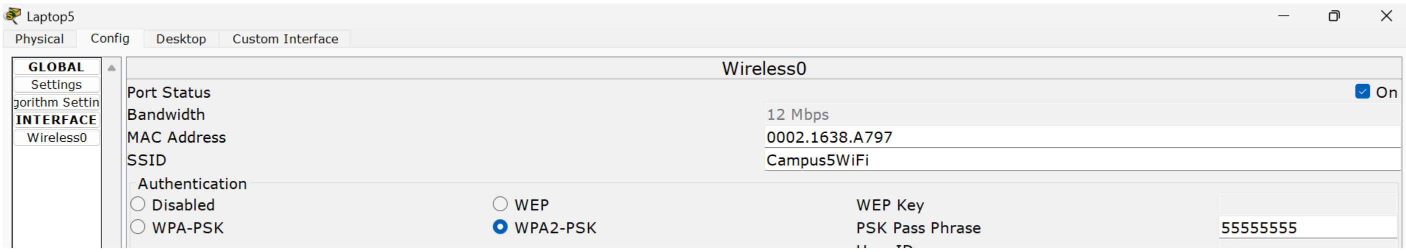
- Wireless connectivity for laptops
- SSID-based access

Wireless access points were configured with SSIDs for each campus.

Examples: campus1wifi, campus2wifi and so on.

Wireless hosts: successfully received IP addresses via DHCP.





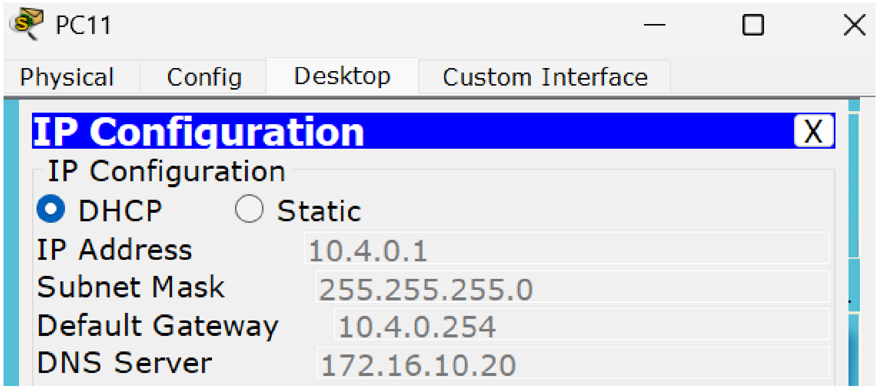
Testing and Verification

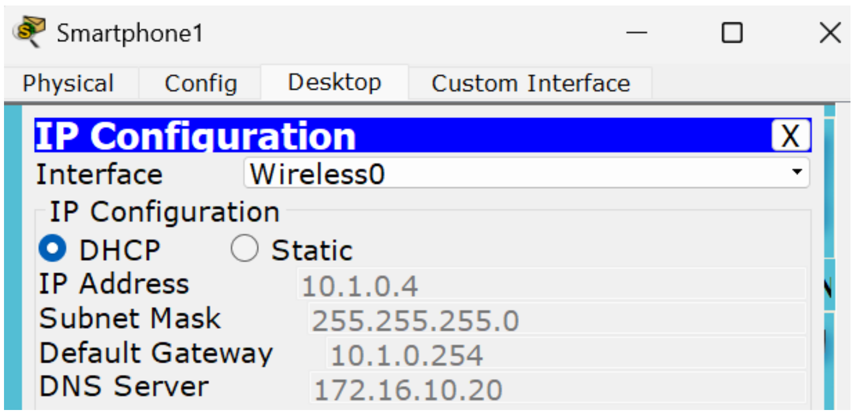
The following tests were performed:

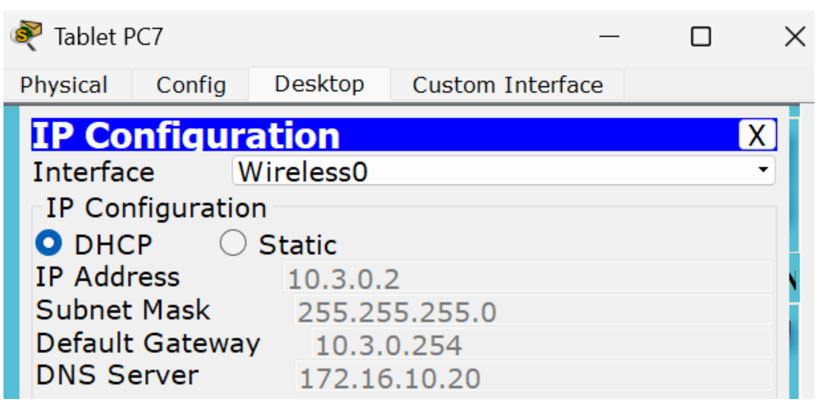
- **Ping Test:** Connectivity between campuses verified.

Fire	Last Statu	Sourc	Destinatic	Type	Colo	Time(s	Period	Num	Edit	Delete
	Failed	Tabl...	PC6	IC...		0.000	N	0	(ed...	(delete)
	Successful	PC6	PC6	IC...		0.000	N	1	(ed...	(delete)

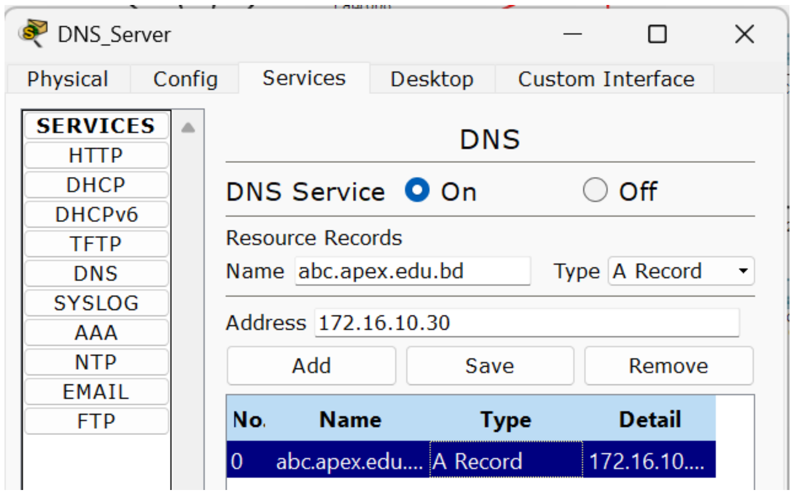
- **DHCP Test:** Automatic IP assignment confirmed on both wired and wireless devices.



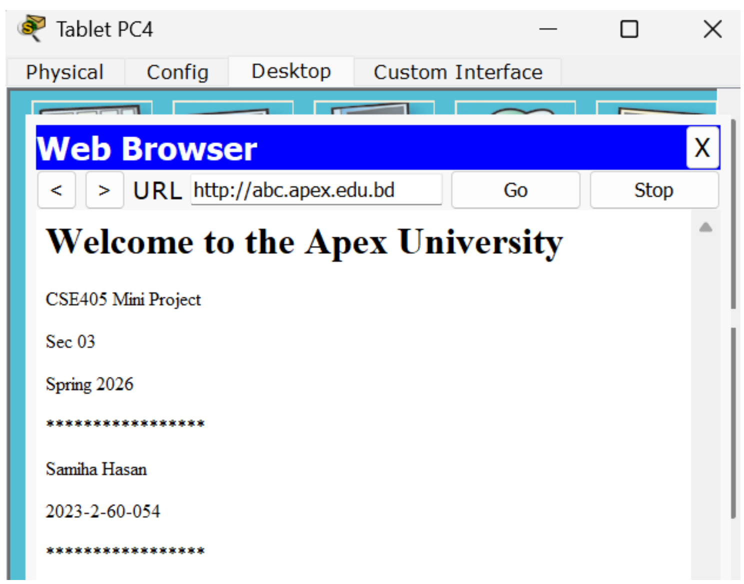




- **DNS Test:** Domain name successfully resolved.



- **Web Access Test:** Website accessed using http://abc.apex.edu.bd



Challenges Faced

Several challenges were encountered:

- IP of three different classes
- DHCP pool configuration
- Wireless connectivity troubleshooting

These issues were resolved through configuration verification and testing.

Limitations & Future Enhancement

The design is limited to simulation within Cisco Packet Tracer.

Real-world deployment would require:

- Advanced security implementation
- Redundancy
- Firewall configuration
- Load balancing

Conclusion

This project successfully implemented a complete university enterprise network. All project requirements were fulfilled including 8 LANs, Dynamic routing using OSPF, DHCP, DNS, Web server and Wireless connectivity. The project demonstrates practical understanding of enterprise-level network design and implementation.

-----------****---------**